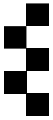
# Third Normal Form (3NF)



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# Third Normal Form (3NF)

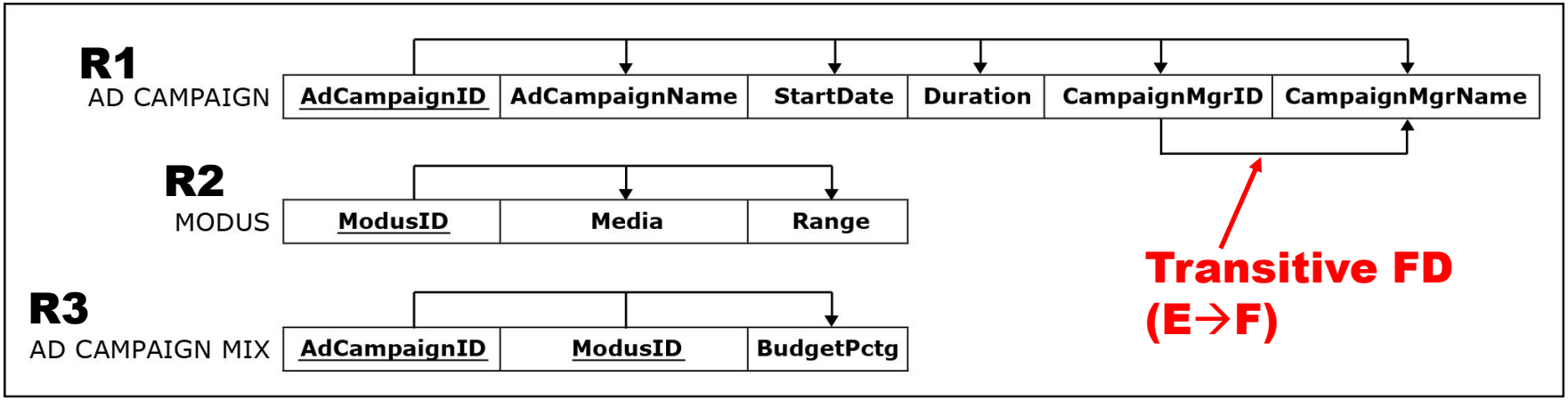
- A relation is in 3NF if it is in 2NF and if it does not contain transitive functional dependencies

# Normalise Relation (AD CAMPAIGN MIX) into 3NF (1)

## Relation R (AD CAMPAIGN MIX)

| A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|
| AdCampaignID | AdCampaign Name | StartDate | Duration | Campaign MgrID | Campaign MgrName | ModusID | Media | Range | Budget Pctg |

## Relation R in Given 2NF:




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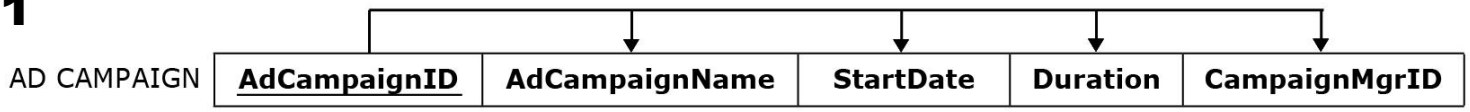
# Normalise Relation (AD CAMPAIGN MIX) into 3NF (2)

- Determine the closure of the determinant of transitive FD (attribute at left side of FD), i.e., $\{E\}^+$=?
- $\{E\}^+$={EF}
- So divide relation R1(A,B,C,D,E,F) into two sub-relations R12(E,F)

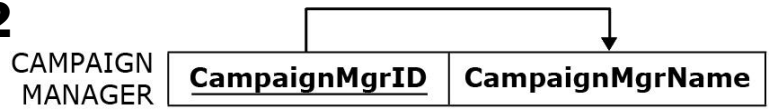
  and

  R11(A,B,C,D,E,F)

**R11**



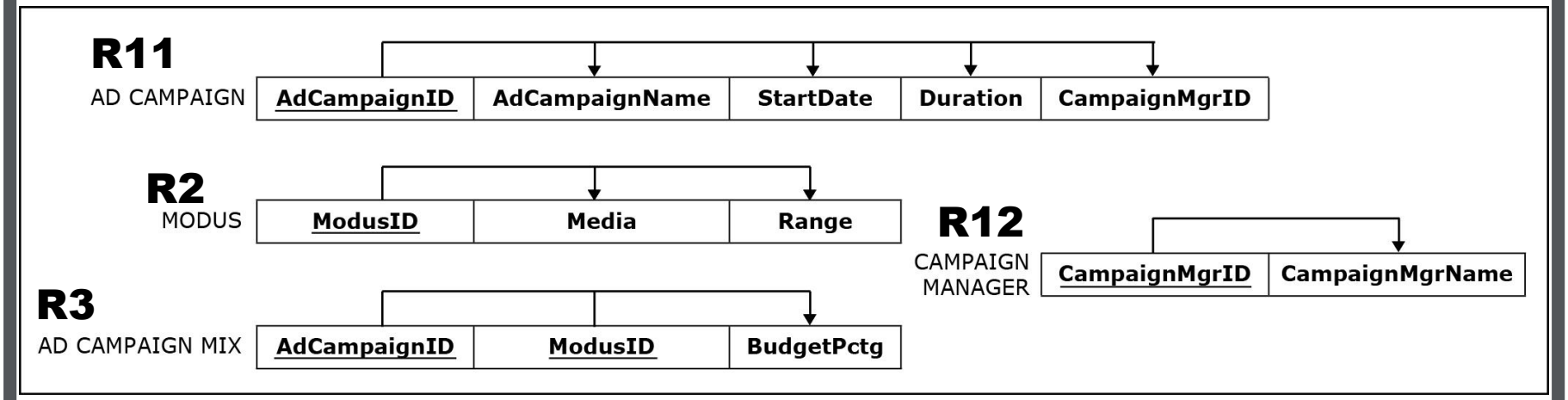

**R12**




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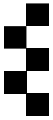
# Normalise Relation (AD CAMPAIGN MIX) into 3NF (3)

Desired 3NF of the relation (AD CAMPAIGN MIX)




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# Boyce-Codd Normal Form (BCNF)



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# Boyce-Codd Normal Form (BCNF)

- A relation is in BCNF if, and only if, it is in 3NF and every determinant is a candidate key.
- BCNF also referred as 3.5 normal form.

## Difference between 3NF and BCNF

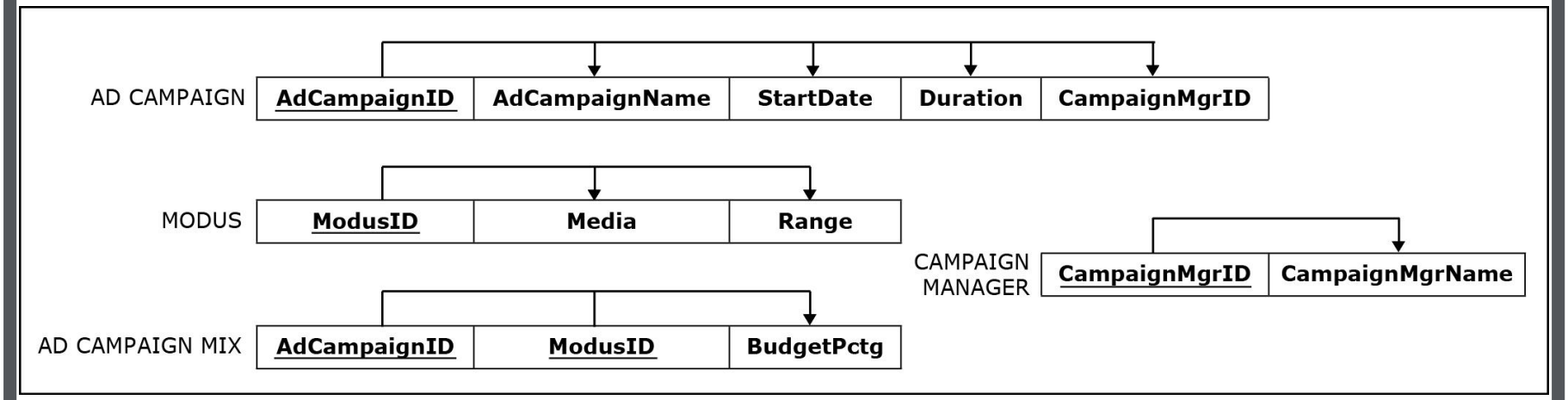
- *non-prime attribute → prime attribute* *(3NF allow such FDs)*
- However, BCNF does not allow such FDs.

## Potential Violation of BCNF

- the candidate keys overlap
  (have at least one attribute in common)

# Normalise Relation (AD CAMPAIGN MIX) into BCNF

## Desired BCNF of relation AD Campaign MIX




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# AD CAMPAING MIX with Update Anomalies

AD CAMPAIGN MIX

| AdCampaignID | AdCampaign Name | StartDate | Duration | Campaign MgrID | Campaign MgrName | ModusID | Media | Range | Budget Pctg |
|---|---|---|---|---|---|---|---|---|---|
| 111 | SummerFun20 | 6.6.2020 | 12 days | CM100 | Roberta | 1 | TV | Local | 50% |
| 111 | SummerFun20 | 6.6.2020 | 12 days | CM100 | Roberta | 2 | TV | National | 50% |
| 222 | SummerZing20 | 6.8.2020 | **30 days** | CM101 | Sue | 1 | TV | Local | 60% |
| 222 | SummerZing20 | 6.8.2020 | **30 days** | CM101 | Sue | 3 | Radio | Local | 30% |
| 222 | SummerZing20 | 6.8.2020 | **30 days** | CM101 | Sue | 5 | Print | Local | 10% |
| 333 | FallBall20 | 6.9.2020 | 12 days | CM102 | John | 3 | Radio | Local | 80% |
| 333 | FallBall20 | 6.9.2020 | 12 days | CM102 | John | 4 | Radio | National | 20% |
| **444** | **AutmnStyle20** | **6.9.2020** | **5 days** | CM103 | Nancy | 6 | Print | National | 100% |
| 555 | AutmnColors20 | 6.9.2020 | 3 days | CM100 | Roberta | 3 | Radio | Local | 100% |
| ???? | ???? | ???? | ???? | ???? | ???? | **7** | **Internet** | **National** | **????** |

**Deletion Anomaly Example:**
Cannot delete campaign 444 without also deleting all the data about campaign manager CM103 and campaign modus 6.

**Modification Anomaly Example:**
To change the duration of the campaign 222 from 30 to 45 days, three records have to be modified.

**Insertion Anomaly Example:**
Cannot insert new campaign modus 7 without inserting an actual campaign using the new modus 7.

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# AD CAMPAING MIX without Update Anomalies

AD CAMPAIGN

| AdCampaignID | AdCampaignName | StartDate | Duration | CampaignMgrID |
|---|---|---|---|---|
| 111 | SummerFun20 | 6.6.2020 | 12 days | CM100 |
| 222 | SummerZing20 | 6.8.2020 | 45 days | CM101 |
| 333 | FallBall20 | 6.9.2020 | 12 days | CM102 |
| 555 | AutmnColors20 | 6.9.2020 | 3 days | CM100 |

**Modification Anomaly Resolved:**
Only one record modified.

CAMPAIGN MANAGER

| CampaignMgrID | CampaignMgrName |
|---|---|
| CM100 | Roberta |
| CM101 | Sue |
| CM102 | John |
| CM103 | Nancy |

MODUS

| ModusID | Media | Range |
|---|---|---|
| 1 | TV | Local |
| 2 | TV | National |
| 3 | Radio | Local |
| 4 | Radio | National |
| 5 | Print | Local |
| 6 | Print | National |
| 7 | Internet | National |

**Deletion Anomaly Resolved:**
Campaign 444 deleted but all the data about the campaign manager CM103 and the campaign modus 6 remain in the database.

AD CAMPAIGN MIX

| AdCampaignID | ModusID | BudgetPctg |
|---|---|---|
| 111 | 1 | 50% |
| 111 | 2 | 50% |
| 222 | 1 | 60% |
| 222 | 3 | 30% |
| 222 | 5 | 10% |
| 333 | 3 | 80% |
| 333 | 4 | 20% |
| 555 | 3 | 100% |

**Insertion Anomaly Resolved:**
New campaign modus 7 inserted without inserting an actual campaign using the new modus 7.

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# Concluding Remarks

So far you have learnt

- Update Operations
- Why redundant data is not good in a relation
- What are update anomalies
- How to find FDs in a relation
- How to find Candidate key(s) in a relation
- What are Full Key FD, partial FD and Transitive FD
- Dependency Preserving Decomposition
- Lossless Join Decomposition
- How to Normalize a relation in 2NF, 3NF and BCNF

## Next Lecture

- Conversion of a relation directly into BCNF
- MVFD, Join Dependencies, 4NF, 5NF
- Security in DBMS (if time permits)

# Suggested Readings

- https://opentextbc.ca/dbdesign01/chapter/chapter-12-normalization/
- https://www.geeksforgeeks.org/lossless-join-and-dependency-preserving-decomposition/
- D1: Database processing: fundamentals, design, and implementation (David M. Kroenke; David J. Auer) . 14th Edition (Global Edition). Pearson.
  - Chapter 3
- The following sections from the textbook Database Systems: A Practical Approach to Design, Implementation, and Management by Connolly and Begg, Global Edition/6th Edition.
  - 14.4 to 14.8, 15.2